

Designing AI-powered search platforms: Advertiser incentives, quality paradox, and AI sequencing

May 25, 2026

Abstract

Search engines face growing competitive pressure from artificial intelligence, which increasingly substitutes for traditional query-based search on platforms. In response, these firms have invested substantially in developing and embedding AI-driven functionalities within their services, but it poses significant challenges to their established sponsored search revenue model. This paper examines a critical tradeoff in platform design: as AI systems become more accurate and helpful, viewers increasingly rely on AI-generated summaries rather than clicking through to advertisers' links. This shift reduces traffic and engagement for advertisers, weakening their incentives to invest in higher-quality content. Paradoxically, stronger AI can undermine content quality by eroding the competitive pressure that drives advertisers to improve.

We uncover four key findings. First, the relationship between AI quality, AI pricing, and advertiser quality is non-monotonic. Improving AI does not always benefit the platform; in fact, higher AI quality can reduce overall profits by cannibalizing sponsored clicks and discouraging advertiser investment. Second, monetizing AI through subscription does not always outperform a free-access strategy. When AI quality is low, free access preserves advertiser incentives and viewer participation, whereas charging for AI reduces engagement and harms content quality. Third, allowing viewers the choice to pay for AI strengthens advertisers' motivation to invest in better content, but the order in which AI is presented relative to sponsored search is a crucial design decision. Finally, the option of bypassing AI service is profitable only when the advertiser's margin from satisfying the viewer is either very high or very low.

Keywords: AI-powered platforms, search engine design, sponsored search, sequencing, revenue management

1 Introduction

Search engines are facing unprecedented competitive pressure from AI systems that deliver direct, synthesized answers, displacing the traditional “ten blue links” paradigm. Commentators increasingly claim that “search is dying” as users route information needs through conversational agents and AI overviews in search results, rather than issuing multiple iterative queries and clicking across publisher pages. Google’s Search Generative Experience (SGE), now AI Overviews, inserts AI-written summaries at the top of results, often above organic listings and sometimes above paid units, a shift widely observed to reduce downstream clicks to websites. Multiple empirical studies report substantial click-through-rate (CTR) declines when AI Overviews appear—including analyses by Ahrefs and industry summaries—as well as Pew Research’s user-behavior study, suggesting growing zero-click behavior and fewer visits to external sites (Law and Guan 2025; Goodwin 2025; Chapekis and Lieb 2025). This threatens a revenue model long anchored in sponsored search, particularly for Google, where advertising has historically comprised roughly 75–80% of total revenue, with the largest share from search and other properties (Statista Research Department 2025; Faria 2026; Alphabet Inc. 2025).

In response, major search platforms have accelerated AI development and **embedded AI overviews/summaries** into their own search experiences. Google’s SGE (AI Overviews) continues to evolve across surfaces and geographies, while Baidu’s ERNIE Bot is integrated into Baidu Search and related services, providing AI-generated summaries, planning assistance, and plugin-based real-time retrieval. These features can scaffold user tasks—for example, producing a first-cut travel plan, decomposing a complicated purchase decision, or organizing constraints for local services—thereby simplifying the subsequent search process even when the AI overview does not fully resolve the query (O’Toole 2023; Baidu Research 2023; Camilla 2025). Industry tracking also finds that AI overviews increasingly occupy prime SERP real estate, altering ad and organic placements and changing where attention flows; several audits document that shopping and text ads sometimes render above or below the AI block, with variability by query class and niche (Agius 2024; Wilt 2024).

However, embedding AI in search introduces tension with the incumbent **sponsored search** model. By satisfying user intent directly, AI summaries can reduce the need for users to click further on organic or sponsored links, potentially diminishing the revenue from search query auctions. While Google, as depicted in Figure 1, claims AI in search is increasing queries and the quality of clicks (statement from Reid (2025), VP & Head of Google Search), other research posits that users are less likely to click on links when an AI summary is displayed. The publishers’ and advertisers’ concerns about reduced traffic available to monetize are further reinforced by recent field data and large-scale keyword studies (Chapekis and Lieb 2025; Law and Guan 2025; Price 2025; Southern 2025); two notable examples are presented in Figure 1 by Pew

research center and Ahrefs. This is consequential because leading search engines operate **two-sided platforms**: they earn on the viewer side (e.g., via subscriptions or emerging AI access models) and on the advertiser side (via sponsored links and related formats). Mechanism-design work highlights that platforms typically trade off rent extraction against click volume and relevance to sustain long-run ad revenue (Gomes 2014; Hoppner 2015). In practice, Google continues to report advertising as the dominant revenue source in SEC filings, while Baidu’s results distinguish online marketing services from other segments—both signaling the materiality of sponsored search to platform economics even as AI capabilities expand (Alphabet Inc. 2025; Statista Research Department 2025; Baidu, Inc. 2025).

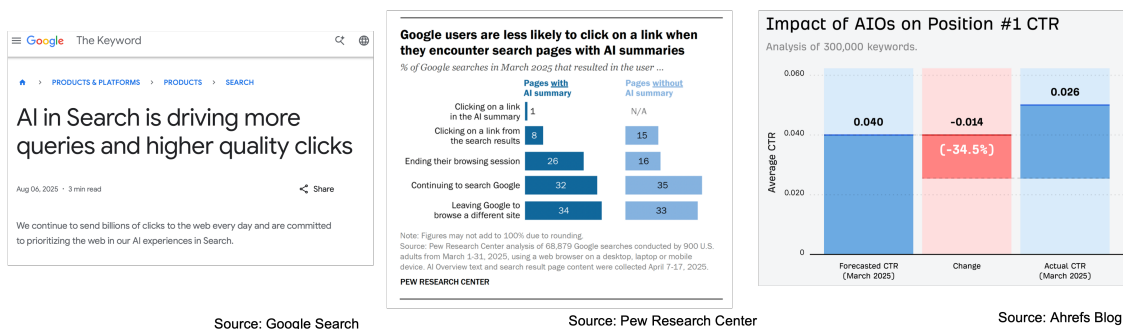


Figure 1: Conflicting Experts’ Opinions: AI Summary Drives More or Fewer Clicks.

This paper goes after all the above conflicts and intends to examine the platform design with AI service, sponsored search, and the two-sided revenue extraction from advertisers and viewers. The analysis begins with a base model in which the AI service is positioned as the first channel, followed by sponsored search. This structure allows us to examine the dual sources of revenue: payments from viewers who access the AI (and search engine) service and payments from advertisers who participate in the sponsored search channel. Within this framework, we compare two distinct pricing schemes. In the first scheme "Sponsored Intelligence" with abundant contemporary examples, the viewer price is set to zero, thereby relying exclusively on advertising revenue. In the second scheme "Subscriber Intelligence," the viewer price is positive; as exemplified by Kagi’s plan in Figure 2, this introduces a direct revenue stream from viewers while potentially influencing demand and advertiser behavior.¹ This comparison provides a foundation for understanding the trade-offs between viewer accessibility and monetization through sponsored search and viewer subscription.

Our analysis reveals a fundamental trade-off between AI quality, advertiser incentives, and platform pricing driven by two distinct forces. First, through a *market expansion* effect, improvements in AI enhance viewer experience and attract more viewers to the platform, yet

1. Kagi is a fully paid search engine, and its core charging model for users is a monthly/annual subscription system (see <https://kagi.com/pricing>).

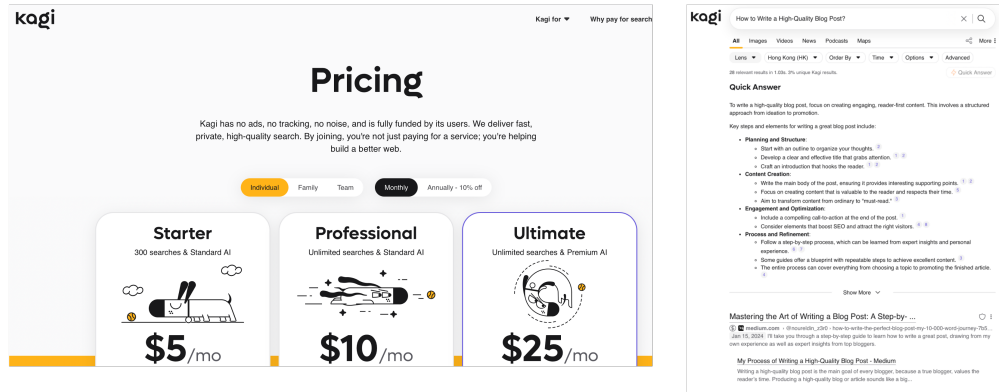


Figure 2: Kagi's Pricing and Web Page Display.

viewers increasingly rely on AI-generated summaries rather than clicking through to sponsored links. Because the decision to click is governed by the quality of the link rather than AI quality per se, improvements in AI alone do not affect the optimal cost-per-click (CPC) when viewer costs are unchanged, while reduced click-through weakens advertisers' incentives to invest in content quality. Second, through a *friction reduction* effect, AI lowers viewers' search and comprehension costs, making viewers more selective and rendering low-quality links unviable. To retain high-quality advertisers under these conditions, the platform must reduce CPC, which raises average content quality but compresses advertising margins. The resulting tension lies between AI-driven market expansion that dampens advertiser incentives and AI-enabled friction reduction that disciplines low-quality content.

We find that the relationship between AI quality, AI pricing, and advertiser quality is **non-monotonic**. As AI quality improves, the advertiser initially raises the quality of their links to satisfy more discerning viewers. Intuitively, higher AI quality should support higher subscription fees. Yet, when AI quality becomes very high, viewers are largely intercepted by front-loaded AI answers, reducing traffic to the advertiser even if they produce superior content. This weakens their incentive to invest further, causing advertiser quality to decline after a peak. At this point, the platform must lower AI subscription fees to attract more viewers and offset the loss of link engagement. Moreover, the platform may **voluntarily give up** monetizing AI through subscription: when AI quality is low, advertiser quality is higher under free access, and charging viewers reduces participation. Conversely, when AI quality is high, free access erodes advertiser quality, making paid AI services more attractive to viewers who prefer efficient AI answers over diminished link content.

Building on the base model, we extend the analysis in two directions. First, we incorporate the option for viewers to bypass the AI service, thereby avoiding the payment of price. We label this as the "**Paywall mechanism**," wherein the platform blocks viewers without payments from the (advanced) AI service. This practice has been implemented by Google and Bing; for

example, a subscription to Google One AI Premium is required for advanced AI features.² This extension allows us to evaluate how the availability of a bypass option alters viewer participation, advertiser exposure, and overall revenue outcomes. Second, we relax the assumption of sequence by allowing the **AI service to follow sponsored search** rather than precede it. This arrangement intriguingly coincides with AI Overview in Google, which sometimes appears after the initial organic search results or ads, rather than at the very top; see Figure 3 for instance.³ This extension enables us to investigate the optimal ordering of the two channels, considering both viewer engagement and advertiser incentives.

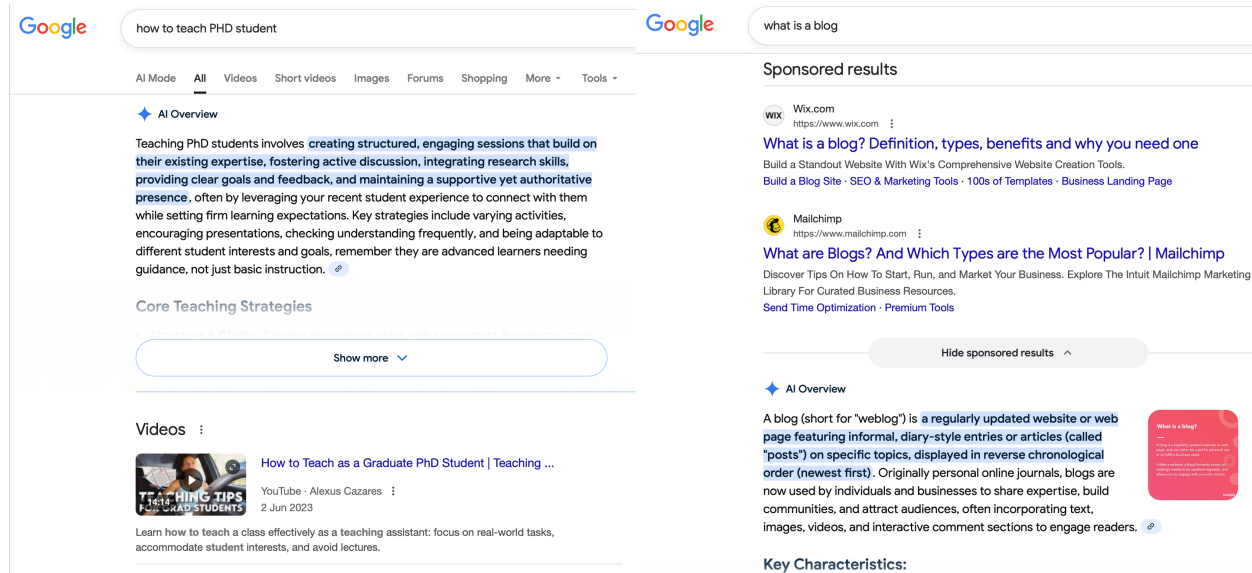


Figure 3: AI Sequencing: Different AI Placements for Google.

Allowing viewers the choice of whether to pay for AI fundamentally **strengthens the advertiser's incentives** to invest in higher-quality content. As AI quality improves, its appeal to paying viewers rises, increasing both their willingness to subscribe and the platform's ability to charge higher prices. This directly boosts platform profits. At the same time, the advertiser benefits from the dual audience: those who click on links after using AI and those who access links without paying. To capture both groups, the advertiser is motivated to raise link quality, which indirectly enhances overall profitability. However, this positive cycle is constrained by costs—if the expense of maintaining high-quality AI becomes excessive, platform profits

2. The paid subscription (\$ 19.99/month) primarily unlocks experimental AI search interfaces, the Gemini Advanced model, and more, while the standard search experience remains unchanged; see https://one.google.com/about/plans?g1_landing_page=0.

3. This shift is part of Google's effort to balance providing instant, synthesized answers with maintaining the traditional, high-traffic "blue link" experience. See <https://www.searchenginejournal.com/google-ai-overviews-show-less-when-users-dont-engage/564694/#:~:text=Stein> and <https://nightwatch.io/blog/ai-overviews/#>.

eventually decline despite higher service quality. In this setting, when viewers must pay before accessing AI, the platform’s ranking strategy is guided by the marginal contribution of AI quality to revenue, balancing subscription pricing against advertiser incentives.

Furthermore, the order in which AI is presented relative to sponsored search is not trivial and should be treated as an endogenous design choice under the Paywall mechanism. Counterintuitively, during the early stages of AI development, it may be optimal to **place AI second**, contrary to current practice that emphasizes showcasing AI capabilities upfront. While leading with AI signals investment and innovation, it may not maximize revenue. If AI quality is high, placing it first highlights its value, encourages willingness to pay, and maximizes direct subscription revenue. Conversely, if AI quality is low, positioning it first risks frustrating viewers and reducing their willingness to pay. In such cases, placing AI second allows the platform to engage viewers initially with higher-quality links, while still offering AI as a value-added option. The ranking strategy thus reflects a trade-off: high-quality AI is prioritized for monetization, while low-quality AI is sequenced later to protect viewer experience and mitigate risks.

Finally, we compare the mechanisms and identify the AI quality as the core factor to facilitate the platform’s optimal arrangement. In the subscriber intelligence model, viewers can either pay for the platform’s services or leave, making high-quality AI crucial for attracting a larger audience. Under the paywall mechanism, cost-per-click becomes a key revenue source, with advertisers’ payoffs impacting AI placement. Low payoffs lead to prioritizing the AI section to encourage payments, while high payoffs incentivize advertisers to improve link quality, resulting in the AI-Second Paywall mechanism. However, this mechanism can be less profitable than the Subscriber Intelligence scheme, where the platform benefits directly from viewers’ willingness to pay for sponsored links. Thus, in between the two extreme scenarios, the Subscriber Intelligence approach ultimately proves more effective than the paywall mechanisms. This **non-trivial pattern of switching** between bypassing and not-bypassing AI showcases how pricing, bypass options, and sequencing interact to shape the equilibrium outcomes of dual-channel revenue models.

The rest of the paper is organized as follows. In Section 2, we review the relevant literature. Section 3 describes the model setting, and Section 4 compares the two schemes with and without viewer subscription fees. Section 5 examines some extended mechanisms and compares across them for platform profitability. Section 6 offers some concluding remarks. All the additional discussions and technical proofs are relegated to the appendix.

2 Literature review

Our research contributes to four distinct streams of literature, highlighting our departure from existing studies.

Sponsored Search. Sponsored search is both a central technology in online advertising and a core focus of academic research. Foundational work by Edelman, Ostrovsky, and Schwarz (2007) and Varian (2007) introduces the generalized second-price (GSP) auction, in which advertisers place per-click bids, positions are assigned in descending bid order, and payments equal the minimum bid needed to secure an ad slot. Subsequent studies enrich the GSP model by incorporating more realistic consumer behavior. For instance, Craswell et al. (2008) document substantial “position bias” in click-through rates (CTRs), showing that ad order is critical. Likewise, Kempe and Mahdian (2008) and Aggarwal et al. (2008) adopt sequential ad-scanning frameworks, characterizing optimal mechanisms under these externalities. In contrast, Balseiro and Désir (2023) moves away from sequential evaluation and examines consumers who click only after reviewing all available options. Compared with this literature, our consumer search process resembles the cascade model, but we extend it by introducing an AI section; and we analyze a two-sided revenue model in which the AI section competes with sponsored links.

Two-Sided Market. The two-sided revenue literature studies mechanisms that jointly govern interactions between advertisers and consumers on a platform. For example, Gomes (2014) examines a monopolistic search engine offering a single sponsored link, where advertisers differ in willingness to pay and consumers differ in clicking costs. The platform uses two-sided transfers: an auction determines the displayed advertiser, while consumers pay (or receive a subsidy) upon clicking. Similarly, Birge et al. (2021) analyze two-sided platform design in a bipartite network, characterizing optimal commission and subscription schemes and highlighting the role of network structure in revenue. Search-theoretic approaches also shed light on click behavior (Athey and Ellison 2011; Chen 2021; Chen and He 2011; Chu, Nazerzadeh, and Zhang 2020). Relative to this literature, we incorporate an AI section and focus on the optimal sequencing between AI and sponsored search. This sequencing creates the possibility for viewers to bypass one of the options—specifically, the AI section—which has no direct counterpart in prior models.

AI Research. The literature on artificial intelligence pertinent to this study centers on AI-generated content. A majority of the literature focuses on AI-generated advertisements and videos (Kapoor and Kumar 2025; Kumar and Kapoor 2024; Exner et al. 2025). Other AI-related studies utilize empirical research and field experiments (Su et al. 2024; Pu, Li, and Cheng 2025). The most relevant literature to our work pertains to AI-generated summaries. Pham, Pham Thi, and Duong (2024) construct a theoretical model of consumer AI-powered information search behavior to identify the main factors affecting consumer search behavior. Yu and Li (2026)

study how AI-generated summaries increasingly satisfy readers’ information needs without rerouting them to original publishers. Their model illustrates that while summaries reduce search costs and can expand readership, substitution pressure may lead top publishers to lower content quality, ultimately harming both summary quality and search outcomes. The study by Liu, Wang, and Liu (2026) includes an analysis of search engines where AI overviews precede sponsored links. They demonstrate a dual effect: these overviews reduce direct clicks on sponsored ads but also escalate competition among advertisers, resulting in increased bids. In contrast to this stream, in our framework AI overviews has market expansion effect to increase the number of viewers who click on the link. More importantly, we allow viewers to bypass the AI section and examine the platform’s *optimal sequencing* of AI and sponsored content.

Platform Operations. The literature related to platform operations addresses scenarios where the platform sells its own private label while also incorporating independent sellers. Long and Amaldoss (2024) employ a game-theoretic model to investigate when platforms prioritize their private labels in sponsored advertising, and how this affects consumers and third-party sellers. The analysis reveals that despite offering private labels at lower equilibrium prices, such self-preferencing ultimately harms consumers. Furthermore, Zou and Zhou (2025) develop an analytical model to evaluate the effects of search neutrality in a scenario where an online platform, acting as a first-party seller, competes with a commission-paying third-party seller. The analysis shows that search neutrality may inadvertently harm both consumers and third-party sellers. The literature on platform operations related to search ranking also includes the work of Feldman et al. (2022), Derakhshan et al. (2022), Cheng et al. (2023), and Long and Liu (2024). Our work relates to the platform operations literature by endogenizing the “recommendation sequence” across different content or advertising options. We embed the cascade model within the viewer’s search process, allowing ordering decisions to interact directly with consumer behavior.

3 Model

To simplify the analysis, we focus on a single sponsored search link. The system therefore consists of one platform, one advertiser, and a continuum of viewers. The platform provides an AI-generated summary section and a single sponsored link offered by the advertiser, with link quality denoted by q_S . The AI-generated summary is presented first and the sponsored link appears subsequently once the viewer has paid (subscribed).

Viewers. Each viewer faces two choices: (i) Pay (subscribe) to access, at price p_V , the AI-generated summary and the sponsored link; or (ii) Do not subscribe, in which case the viewer abstains from the platform’s services. Conditional on subscribing, the interaction proceeds as follows. If the viewer finds the AI summary satisfactory (with probability q_A), they exit without

clicking the sponsored link. If the AI summary is unsatisfactory (with probability $1 - q_A$), the viewer proceeds to the sponsored link and may click if expected benefits exceed costs.

The utility of viewer i from subscribing is

$$U_i^1 = \max \{0, q_A(v - p_V) + (1 - q_A) U_i^2\}. \quad (1)$$

In equation (1), 0 is the normalized value from no subscription; if the viewer subscribes to the platform's service, with probability q_A , she will be satisfied with the AI section and exit. Here, v is the valuation of content, and p_V is the subscription price (for AI and access to the sponsored link). For the remaining probability $1 - q_A$, she enters the sponsored link stage, and obtains the continuation value U_i^2 :

$$U_i^2 = \max \{ -p_V, q_S v - p_V - \theta_i / Q \}. \quad (2)$$

In Equation (2), θ_i is viewer i 's intrinsic search cost that is uniformly distributed over $[0, 1]$. If in the sponsored link stage, the viewer decides not to click, her net payoff is $-p_V$. When she clicks, her expected payoff from the content is $q_S v$, and clicking comes with the modified search cost θ_i / Q . We let the effective search-cost shifter be

$$Q = 1 + b q_A, \quad (3)$$

where b is the *cost-reduction parameter* capturing how AI quality lowers search frictions. This formulation implies that AI summary lowers the "friction" – search cost – traditionally involved in scanning and comprehending sponsored results. The cost reduction induced by AI arises from two channels. First, the AI summary performs initial information processing and synthesis, reducing viewers' cognitive load (mental effort) needed to understand or evaluate sponsored ads/links. Second, the AI summary substitutes for (or subsidizes) the viewer's own cost of acquiring and interpreting information from sponsored results. In both cases, having AI summary prior to the sponsored search reduces the cognitive and informational burden on the viewer's side, and improves the efficiency of viewer-advertiser matching that potentially benefits the advertiser.

The assumption of a homogeneous valuation v is consistent with the classical cascade model in the computer science and economics literature, where viewers derive value primarily from whether their need is satisfied by the content; see Athey and Ellison (2011) that first rationalize the micro-foundation of cascade model with costly search. In our setting, this valuation reflects the intrinsic usefulness of the content itself and is therefore largely independent of the artificial intelligence system facilitating the search. In contrast, search costs are heterogeneous across viewers. The parameter θ_i captures differences in opportunity cost of time, familiarity

with the search environment, prior knowledge of relevant keywords or sources, and the cognitive effort required to process and evaluate search results. These factors naturally vary across individuals, making heterogeneity in search costs a more realistic modeling choice.

The quality levels q_A , q_S correspond to how well the platform's own AI development and the sponsored advertiser's link can match the viewer's need for a specific query. Given (1)–(3), the mass of viewers who subscribe is

$$M_1 = \frac{q_A v - p_V + (1 - q_A) q_S v}{(1 - q_A) \theta_c / Q}, \quad (4)$$

where θ_c is the cutoff search cost implied by the AI summary (scaled by Q per (3)).

Advertiser's Utility. Let M_2 denote the number who click the sponsored link (among subscribers), and N_2 denote the number of subscribers who click the sponsored link and find it satisfactory. From the flow and (4),

$$M_2 = (1 - q_A) M_1 - \max\{0, \frac{q_A v(1 - q_S) - p_V}{\theta_c / Q}\}, \quad (5)$$

$$N_2 = q_S M_2, \quad (6)$$

where $\max\{0, \frac{q_A v(1 - q_S) - p_V}{\theta_c / Q}\}$ denotes the number of viewers who choose not to click the sponsored link, as illustrated in Figure 4. When the AI price satisfies $p_V < q_A v(1 - q_S)$, viewers may still obtain positive utility even without clicking the sponsored link. In contrast, when $p_V > q_A v(1 - q_S)$, all viewers who are dissatisfied with the AI must rely on clicking the link to achieve positive expected utility.

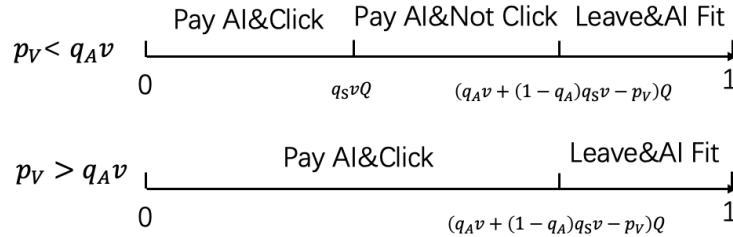


Figure 4: Viewer's Distribution.

The advertiser earns α per satisfied viewer and incurs cost-per-click. With quadratic link-quality cost $C_2 = c q_S^2$, the advertiser's utility is

$$V = \alpha N_2 - p_S M_2 - C_2 = (\alpha q_S - p_S) M_2 - c q_S^2, \quad (7)$$

where (6) substitute into the first expression.

Platform's Objective. The platform maximizes expected revenues from viewers and the

Symbol	Definition
q_A	AI summary quality; probability viewer is satisfied by AI.
q_S	Sponsored link quality; probability viewer is satisfied by link.
b	Effect of AI quality on reducing search cost.
Q	Search-cost shifter: $Q = 1 + b q_A$.
v	Viewer's valuation of content.
p_V	Subscription price for AI and sponsored link.
θ_i	Viewer i 's intrinsic search cost.
θ_c	Cutoff search cost for subscription.
U_i^1	Utility from subscribing (AI first, then link if needed).
U_i^2	Utility from sponsored link stage.
M_1	Mass of viewers who subscribe.
M_2	Subscribers who click link: $M_2 = (1 - q_A) M_1$.
N_2	Subscribers satisfied by link: $N_2 = (1 - q_A) q_S M_1$.
α	Advertiser payoff per satisfied viewer.
p_S	Cost-per-click paid by advertiser.
c	Advertiser's cost coefficient for link quality.
C_2	Advertiser's cost: $C_2 = c q_S^2$.
V	Advertiser's utility: $V = \alpha N_2 - p_S M_2 - C_2$.
k	Platform's cost coefficient for AI quality.
Π	Platform profit: $\Pi = M_1 p_V + p_S M_2 - k q_A^2$.

Table 1: Notation table.

advertiser net of AI costs. Using (4) and (5), the profit function is

$$\Pi = M_1 p_V + p_S M_2 - k q_A^2. \quad (8)$$

In (8), the first term is subscription revenue, the second is click revenue from the advertiser, and the last term is the convex cost of providing AI quality. Table 1 summarizes the key notation used in this paper, and Figure 5 illustrates the sequential decision making process of each viewer.

The primary goal of such a theoretical model is to understand the fundamental shift introduced by the AI summary's presence. By simplifying the sponsored link landscape to a single entity, our model can isolate and clearly analyze the direct impact of the AI summary on viewer behavior. We can also demonstrate how the AI summary can act as an information filter, and examine the corresponding effect on the overall platform revenue. As we shall see subsequently, the perceived scarcity of traffic or enhancement of viewer engagement will likely occur even if we extrapolate the model to accommodate multiple advertisers' competition.

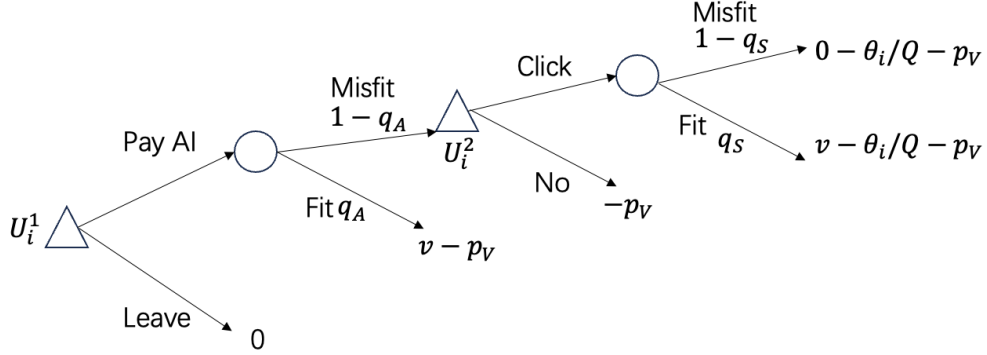


Figure 5: Viewer's Decision Tree.

4 Equilibrium analysis

We now carry out the equilibrium outcomes in the two schemes. We will then compare them and examine the platform profitability.

4.1 Sponsored Intelligence: AI Price $p_V = 0$

In this section, we examine the Sponsored Intelligence scheme in which the AI component is provided at no cost to viewers; hence, $p_V = 0$. Accordingly, all viewers first consume the AI-generated content prior to any subsequent interaction. The timing is as follows. In the first stage, the platform selects the cost-per-click (CPC), denoted p_S . In the second stage, given p_S , the advertiser chooses q_S . Afterwards, viewers' search takes place.

Lemma 1. *Under the Sponsored Intelligence scheme, with exogenous AI quality, the advertiser sets the optimal quality as $q_S^* = 1$.*

For $b = 0$, the equilibrium CPC, the number of viewer who pay AI and platform's profit are

$$\begin{aligned} p_S^{FA0*}(q_A) &= \frac{\alpha v - c\theta_c}{v}, \\ M_1^{FA0*}(q_A) &= \frac{v}{\theta_c(1 - q_A)}, \\ \pi^{FA0*}(q_A) &= \frac{v\alpha}{\theta_c} - c - kq_A^2, \end{aligned}$$

and the number of viewer who pay AI increases in AI quality q_A , platform's profit decreases in AI quality q_A .

For $b > 0$, the equilibrium CPC, the number of viewer who pay AI and platform's profit are

$$\begin{aligned} p_S^{FA0*}(q_A) &= \frac{\alpha v(1 + bq_A) - c\theta_c}{v(1 + bq_A)}, \\ M_1^{FA0*}(q_A) &= \frac{v(1 + bq_A)}{\theta_c(1 - q_A)}, \\ \pi^{FA0*}(q_A) &= \frac{v\alpha(1 + bq_A)}{\theta_c} - c - kq_A^2. \end{aligned}$$

The optimal CPC increases in q_A , α , b . The optimal number of viewers increases in q_A , b . The optimal platform's profit is increasing in α, b and non-monotonic in q_A , specifically, it increases in AI quality q_A when q_A is small; it decreases in AI quality q_A when q_A is large.

Lemma 1 characterizes the equilibrium outcomes with exogenous AI quality and highlights several key economic forces in the search platform. First, the introduction of AI increases overall platform participation by attracting more viewers. From the advertiser's perspective, the key challenge is to retain these viewers and convert them into clicks. As a result, advertisers optimally raise the quality of their sponsored links to the highest level in order to maximize engagement and conversion. Second, AI has two effects: the first is *market expansion effect*, which brings more viewers onto the platform; the second is *friction reduction effect*, which lowers the cost for viewers to click on a link. Therefore, the number of viewers increases with AI quality. Higher-quality AI improves the user experience and lowers effective search frictions, which makes the platform more attractive and draws in more viewers, regardless of whether explicit cost-reduction is present.

Third, AI quality does not affect the optimal CPC in the absence of cost reduction. While better AI attracts more viewers to the platform, the decision to click ultimately depends on the quality of the sponsored link rather than the AI itself. However, when AI also reduces costs, the optimal CPC becomes increasing in AI quality. Intuitively, higher AI quality raises the platform's operating costs, and the platform responds by charging advertisers a higher CPC to offset these costs. Fourth, both the optimal CPC and the number of viewers increase with the cost-reduction parameter. When AI lowers viewers' search and evaluation costs, it enhances their ability to process information and identify relevant content. This leads to greater platform participation and more traffic being directed to sponsored links, which in turn allows the platform to charge a higher CPC.

Finally, when the AI part is free, it is straightforward that the platform's profit increases with the cost-reduction parameter and the strength of network effects. As advertiser quality improves and more viewers join the platform, the resulting expansion in engagement and advertiser willingness to pay leads to higher platform profits. More interestingly, the optimal platform profit decreases in AI quality in the absence of cost reduction and exhibits a non-monotonic relationship with AI quality when AI also reduces cost. Without the friction

reduction effect, the platform can only profit from viewers, yet because AI is free under this mechanism, it has no incentive to enhance its AI quality. However, with friction reduction effect from AI, when AI quality is low, profit rises as improvements in AI quality attract more viewers and enable the platform to increase CPC, thereby enhancing revenue. Conversely, when AI quality is high, incremental improvements reduce profit because the AI cost is too high.

Next, the platform chooses the optimal quality. The timing is as follows. In the first stage, the platform selects the cost-per-click (CPC), denoted p_S , and AI quality, denoted q_A . In the second stage, given p_S , the advertiser chooses q_S . Afterwards, viewers' search takes place.

Lemma 2. *Under the Sponsored Intelligence scheme, with endogenous AI quality, for $b = 0$, the equilibrium AI quality, CPC, the number of viewer who pay AI and platform's profit are*

$$\begin{aligned} q_A^* &= 0, \\ p_S^{FA0*} &= \alpha - \frac{\theta_c}{v}, \\ M_1^{FA0*} &= \frac{v}{\theta_c}, \\ \pi^{FA0*} &= \frac{v\alpha}{\theta_c} - c. \end{aligned}$$

For $b > 0$, the equilibrium AI quality, CPC, the number of viewer who pay AI and platform's profit are

$$\begin{aligned} q_A^* &= \frac{bv\alpha}{2k\theta_c}, \\ p_S^{FA0*} &= \alpha - \frac{2kc\theta_c^2}{v(2k\theta_c + \alpha vb^2)}, \\ M_1^{FA0*} &= \frac{v(b^2v\alpha + 2k\theta_c)}{\theta_c(2k\theta_c - bv\alpha)}, \\ \pi^{FA0*} &= \frac{v\alpha(4k\theta_c + b^2v\alpha)}{4k\theta_c^2} - c. \end{aligned}$$

The optimal AI quality, CPC, the optimal number of viewers and platform's profit increase in α, b .

Lemma 2 shows that when AI quality is endogenously chosen, the platform's incentives to invest in AI depend critically on whether AI generates cost-reduction benefits. If AI is provided for free and does not reduce costs, improving its quality only raises the platform's investment cost without creating additional revenue, so the platform optimally chooses not to invest in AI quality. In contrast, when AI delivers cost-reduction benefits, the platform has a clear incentive to improve AI quality. Higher-quality AI enhances the user experience and lowers effective search frictions, which attracts more viewers to the platform. The resulting increase in traffic raises the value of advertising opportunities, allowing the platform to charge a higher CPC. Therefore, the stronger the cost-reduction effect of AI, the greater the platform's incentive to

invest in its quality.

4.2 Subscriber Intelligence: AI Price $p_V > 0$

In this section, we allow the platform to charge the viewer and examine the two-sided revenue model. The platform first determines the cost-per-click (CPC), after which the advertiser selects the sponsored-search intensity q_S . Subsequently, the platform sets the AI price p_V . The advertiser participates in the platform only if its profit remains non-negative.

Lemma 3. *Under the Subscriber Intelligence scheme, the platform sets the optimal p_S^{SA*} such that the advertiser breaks even.*

Lemma 3 shows that once the platform begins charging the viewer for access to its AI service, it will simultaneously push the advertiser's payment to the highest feasible level—specifically, setting the cost-per-click (CPC) as high as possible subject to the advertiser breaking even. This reflects the platform's two-sided pricing power: by monetizing the viewer directly, it can extract greater surplus from the advertiser, knowing that the value of traffic is enhanced by AI's role in guiding user decisions.

Lemma 4. *Under the Subscriber Intelligence scheme, for the special case $b = 0$,*

1. *the optimal price p_V^{FA*} is increasing in AI quality q_A ;*
2. *the optimal advertiser's quality q_S^{FA*} is increasing in AI quality q_A ;*
3. *the optimal CPC p_S^{FA*} is increasing in AI quality q_A .*

Lemma 4 addresses the special case where AI does not reduce friction in link-clicking. In this setting, the AI price naturally rises with AI quality, since higher-quality AI generates greater willingness to pay from the viewer. At the same time, because AI does not lower the cost of clicking through, the advertiser must compete more aggressively to retain the viewer's attention. As AI quality improves, sponsored link quality also rises, ensuring that the viewer remains engaged even when AI answers are available. This dynamic elevates the value of traffic, allowing the platform to increase CPC and capture more profit from the advertiser.

Proposition 1. *Under the Subscriber Intelligence scheme, for $b > 0$, p_S^{FA*} is increasing in b and q_A , p_V^{FA*} and q_S^{FA*} are increasing in b and non-monotonic in q_A , specifically, they increase in AI quality q_A when q_A is small and decrease in AI quality q_A when q_A is large. Π^{FA*} is increasing in b and q_A .*

Proposition 1 implies that when the platform begins charging viewers for its AI part, the pricing instruments, quality investments, and the resulting market dynamics change substantially. First, the cost-reduction parameter raises CPC, the AI price, and advertiser quality. A substantial reduction in effective click costs indicates that the AI layer delivers decision-making

efficiency to viewers, which strengthens the platform's pricing power on both sides of the market. The platform can therefore charge a higher price for AI access on the viewer side while simultaneously increasing CPC on the advertiser side, reflecting the enhanced value of traffic. In parallel, advertisers invest in higher-quality content to attract and satisfy the expanding viewer base. Second, CPC increases with AI quality. As the quality of the AI layer rises, viewers' search costs fall and adoption expands, which augments the expected returns from sponsored placements. The platform is thus able to set a higher CPC, as advertisers benefit from improved matching and a larger, more engaged audience.

Third, the AI price and advertiser quality are non-monotonic in AI quality. At low to moderate levels of AI quality, advertisers raise link quality to meet higher viewer expectations, and the platform can charge more for AI access. However, once AI quality becomes sufficiently high, a large share of viewer demand is satisfied by the front-loaded AI answers. Even high-quality links then receive less incremental traffic, weakening advertisers' production incentives and causing advertiser quality to decline beyond a peak. To mitigate the resulting demand shortfall and sustain participation, the platform reduces the AI price to attract additional viewers and offset the loss stemming from diminished link quality. Fourth, the platform's profit increases with AI quality. As AI quality improves, the platform can raise CPC and, up to the participation constraint, increase the AI price, while attracting more viewers. The combined effects—higher monetization rates and larger audience size—ultimately yield greater platform profit.

4.3 Comparison between Sponsored and Subscriber Intelligence schemes

In this section, we analyze the platform's profits and investigate whether charging viewer subscription is beneficial.

Lemma 5. *The advertiser's quality, CPC and the number of viewers under the Sponsored Intelligence scheme are higher than that under the Subscriber Intelligence scheme.*

When there is no subscription fee for the platform service, the lower access cost expands the viewer base, increasing total attention on the platform. However, introducing a viewer-side payment for the platform service does not necessarily raise advertiser quality. The reason is that charging viewers screens out marginal users and can reduce click-through opportunities, thereby weakening advertisers' incentives to invest in high-quality links when the incremental traffic reward diminishes.

Proposition 2. *The platform charges the viewers subscription fees when AI quality is high $q_A > \hat{q}_{A1}^{FA}$, and offers the service for free when AI quality is low $q_A < \hat{q}_{A1}^{FA}$.*

Proposition 2 further develops upon Lemma 5 and demonstrates that monetizing the platform service on the viewer side does not universally dominate a no-subscription strategy.

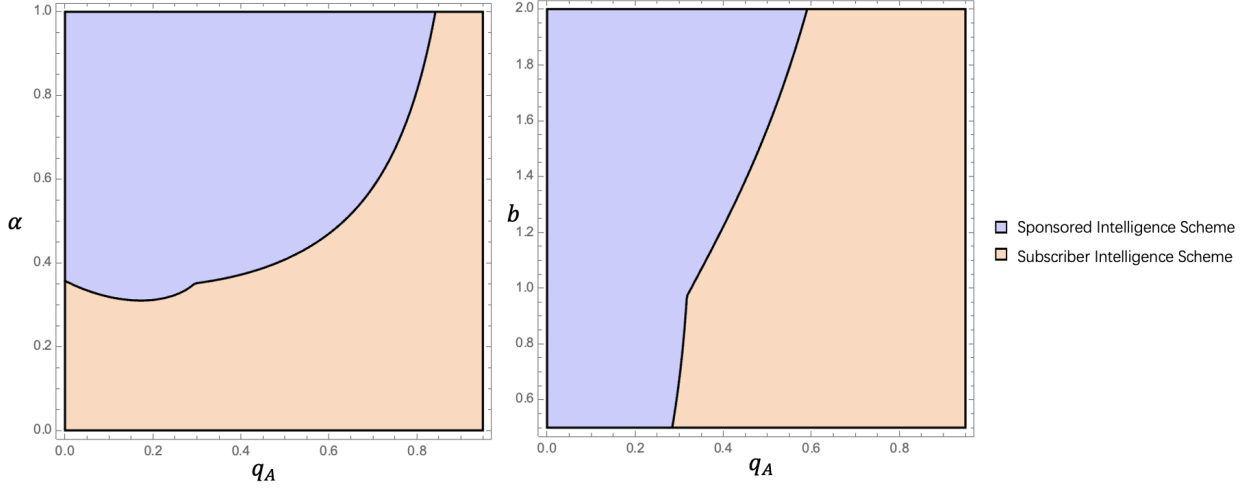


Figure 6: Comparison between Sponsored and Subscriber Intelligence Schemes ($b = 1.5$)

When the underlying AI quality is low, viewers' search frictions remain relatively high and the incremental value of the service is limited; in this region, maintaining no subscription fee supports a larger audience and—by preserving traffic—encourages higher advertiser quality than under a paywall. By contrast, when AI quality is high, the front-loaded answers satisfy a substantial portion of viewer demand, which can markedly reduce the marginal traffic available to publishers if access remains frictionless. In this case, charging for the platform service becomes more attractive: from the viewer perspective, paying for the service substitutes for further link exploration, and from the platform perspective, the higher perceived value justifies pricing while mitigating excessive free-riding on publisher content. As a result, at high AI quality, a subscription (or per-use) fee can restore balance by monetizing the enhanced value delivered to viewers even as advertisers face weaker incremental traffic from clicks.

5 Extensions

We extend the based model in a number of dimensions in the sequel.

5.1 Paywall mechanism: Option for viewers to bypass the AI service

In this section, we incorporate the option for viewers to bypass the AI service, thereby avoiding the payment of price. Namely, viewers first decide whether to pay for AI. Only after making the payment does the possibility of seeing the AI's location become available. Conversely, if the viewer chooses not to pay, they will be categorically denied access to the AI component. This sequential structure captures how the AI layer screens demand and reallocates clicks.

Paying grants access to an AI-generated section presented up front. With probability q_A , the viewers are satisfied and leave; with probability $(1 - q_A)$, they proceed to click the second link, where their probability of satisfaction becomes q_S . If they choose not to pay, they do not see the AI section and directly click the link.

Recall that we allow for heterogeneous search costs across viewers. The utility of viewer i is

$$U_i^0 = \max\{U_i^1, U_i^3, 0\}, \quad (9)$$

where the payoff from paying for AI and following the two-stage path is

$$U_i^1 = \max\{0, q_A(v - p_V) + (1 - q_A)U_i^2\}, \quad (10)$$

where the continuation payoff after an unsatisfactory AI outcome (still having paid) is

$$U_i^2 = \max\{-p_V, q_S v - p_V - \theta_i/Q\}. \quad (11)$$

The interpretations of U_i^1 and U_i^2 have been offered previously.

The new component U_i^3 , corresponds to the payoff from bypassing AI and clicking links directly:

$$U_i^3 = \max\{q_S v - \theta_i, 0\}. \quad (12)$$

In such a scenario, the viewer evaluates the expected payoff from the sponsored search, and weighs between the (intrinsic) search cost θ_i and the anticipated benefit $q_S v$. Compared with the base model, the viewer faces more flexible choices of such subscribing, clicking, and searching behaviors. In the following, we will reexamine the sequential decision making among the viewers, the advertiser, and the platform. Figure 7 illustrates the viewer's dynamic search process under the Paywall mechanism, and Figure 8 describes the endogenous segmentation among different viewers.

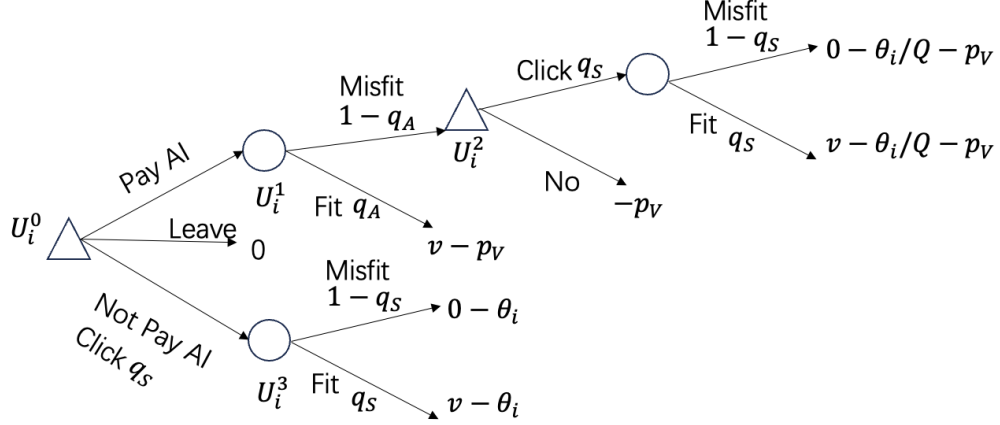


Figure 7: Viewer's Decision Tree in AI-First Paywall mechanism.

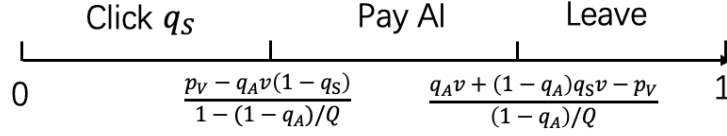


Figure 8: Viewers' Distribution in AI-First Paywall mechanism.

The number of viewers who decide to pay for AI is

$$M_1 = \frac{q_A v + (1 - q_A)q_s v - p_V}{(1 - q_A)\theta_c/Q} - \frac{p_V - q_A v(1 - q_s)}{(1 - (1 - q_A)/Q)\theta_c},$$

and the number of viewers who click the link is

$$M_n + M_2 = \frac{p_V - q_A v(1 - q_s)}{(1 - (1 - q_A)/Q)\theta_c} + (1 - q_A)M_1 - \max\{0, \frac{q_A v(1 - q_s) - p_V}{\theta_c/Q}\},$$

where $M_n = \frac{p_V - q_A v(1 - q_s)}{(1 - (1 - q_A)/Q)\theta_c}$ is the number of viewers who choose not to pay for AI and click the link directly, $M_2 = (1 - q_A)M_1 - \max\{0, \frac{q_A v(1 - q_s) - p_V}{\theta_c/Q}\}$ is the number of viewers who pay AI and click the link.

With these expressions, the platform's profit can be written as

$$\Pi = p_V M_1 + p_S (M_n + M_2) - k q_A^2,$$

which reflects three components: direct subscription revenue from AI-paying viewers, advertising revenue from both non-paying viewers and those who pay but still click links, and the

quadratic cost of improving AI quality. In parallel, the advertiser's profit is given by

$$V = \alpha q_S(M_n + M_2) - p_S(M_n + M_2) - cq_S^2,$$

which captures the value generated from sponsored link clicks, net of the CPC payments to the platform and the cost of enhancing sponsored link quality.

Since the AI section precedes the sponsored link, we call this AI-First Paywall mechanism. Note that if $p_V = 0$, this mode will reduce to the Sponsored Intelligence Scheme. Going beyond this special case, AI quality and pricing strategies jointly shape equilibrium outcomes in this two-sided market. Use "FAw" (in the reverse order) as the superscript to highlight this arrangement.

Proposition 3. *With the option to bypass AI in the Paywall mechanism:*

- AI access fee p_V^{FAw*} , advertiser's quality q_S^{FAw*} and CPC p_S^{FAw*} are increasing in AI quality q_A .
- The profit of the platform is increasing in AI quality q_A if $k < \bar{k}$; otherwise, it increases in q_A when q_A is small, and it decreases in q_A when q_A is large.

Proposition 3 indicates that when viewers can choose whether to pay for the AI, the quality of the AI has a positive impact on the advertiser's quality, AI pricing, and platform profit. The enhancement of AI quality directly amplifies its appeal to paying viewers, elevating both their willingness to pay and the price point of AI services, thereby generating a direct positive effect on platform profits. Meanwhile, to better convert both viewers who click on links after paying for AI and those who access links for free, the advertiser is incentivized to enhance link quality, which in turn indirectly contributes to the overall profitability of the platform. However, if the cost of high-quality AI is too high, once the platform quality reaches a certain level, platform profit will decline due to the excessive cost.

5.2 Allowing the AI service to follow sponsored search

In this section, we analyze the scenario in which the sponsored link is shown before the AI section. This ordering changes the dynamics of the viewer's decision process. Similar to the previous extension, the viewer must first decide whether to pay for the AI service. If the viewer chooses to purchase AI, they are initially exposed to the sponsored link. With probability q_S , the sponsored link satisfies the viewer, and the search ends. With probability $(1 - q_S)$, however, the viewer remains dissatisfied and proceeds to consult the AI section, where the probability of satisfaction becomes q_A .

By contrast, if the viewer decides not to purchase AI, they directly see the sponsored link, and the search terminates without the option of accessing the AI section. This sequencing highlights the trade-off between monetizing AI directly and relying on sponsored content to

retain viewer engagement. To distinguish, we label the mechanism in the previous section as AI-First Paywall mechanism, and the new arrangement as "**AI-Second** Paywall mechanism." Formally, the utility function of viewer i is defined as

$$U_i^0 = \max\{U_i^1, U_i^3, 0\}, \quad (13)$$

where

$$U_i^1 = \max\{0, q_S(v - p_V - \theta_i) + (1 - q_S)U_i^2\}, \quad U_i^2 = \max\{-p_V, q_A v - p_V - \theta_i\}, \quad U_i^3 = \max\{0, q_S v - \theta_i\}.$$

Compared with the previous section, two options are identical: leaving the platform directly yields reservation utility 0, and U_i^3 corresponds to the case where the viewer does not pay for AI and only evaluates the sponsored link. The departure emerges from the option of paying for the AI service U_i^1 . Under the AI-Second Paywall mechanism, U_i^1 represents the case in which the viewer pays for AI and may be satisfied either by the sponsored link or, if dissatisfied, by the AI section. U_i^2 captures the fallback utility when the viewer pays for AI but relies primarily on the AI's answer. The swapping of the sequence factors into the viewer's payoff calculations. The equilibrium choice depends on the subscription price p_V . Specifically,

$$U_i^0 = U_i^1 \quad \text{if } p_V < q_A v(1 - q_S), \quad U_i^0 = U_i^3 \quad \text{if } p_V > q_A v(1 - q_S).$$

Figure 9 demonstrates the viewer's sequential decision making process under this AI-Second Paywall mechanism.

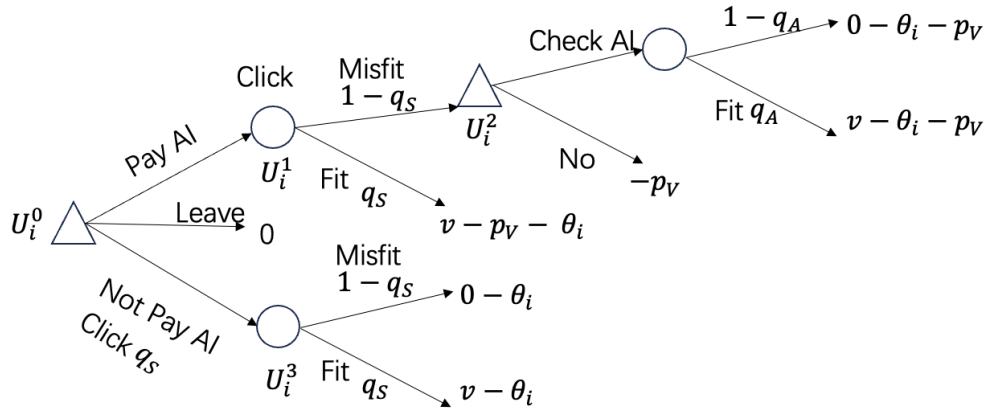


Figure 9: Viewer's Decision Tree in AI-Second Paywall mechanism.

To ease illustration, we note that the platform always get more profits under $p_V < q_A v(1 - q_S)$ (see Proof of Proposition 4 in the appendix), and we focus on this price regime below. The

number of viewers who pay for AI is

$$M_1 = \frac{v(q_S + q_A(1 - q_S)) - p_V}{\theta_c},$$

and the number of viewers who click on the sponsored link is identical, $M_2 = M_1$.

Intuitively, when the AI price is relatively low, viewers find it worthwhile to purchase AI, since they benefit both from the AI's assistance and the possibility of being satisfied by sponsored links. This dual channel of satisfaction increases engagement and generates revenue for both the platform and the advertiser. As indicated in Figure 10, the viewer segmentation is quite different from that under the AI-First Paywall mechanism.

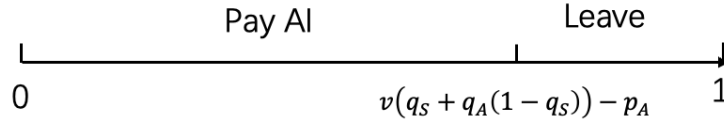


Figure 10: Viewers' distribution in AI-Second Paywall mechanism.

The profit function of the platform can then be expressed as

$$\Pi = p_V M_1 + p_S M_2 - k q_A^2, \quad (14)$$

which consists of subscription revenue from AI-paying viewers, advertising revenue from link clicks, and the quadratic cost of improving AI quality. Similarly, the profit of the advertiser is

$$V = \alpha q_S M_1 - p_S M_1 - c q_S^2, \quad (15)$$

which reflects the value generated from sponsored link clicks, net of CPC payments to the platform and the cost of enhancing sponsored link quality. Finally, when the condition $c\theta_c > (1 - q_A)v\alpha$ is satisfied, the optimal CPC p_S^* is determined by the break-even condition $V = 0$. The superscript "SAw" (in the reverse order) depicts the AI-Second Paywall mechanism.

Proposition 4. *Under AI-Second Paywall mechanism, the profit of the platform is increasing in AI quality q_A if $k < \tilde{k}$. Otherwise, it increases in q_A when q_A is small, and it decreases in q_A when q_A is large.*

Proposition 4 indicates that when viewers can choose whether to pay for AI and the platform places the AI section afterward, the quality of AI has a positive effect on the advertiser's quality, AI pricing, and CPC. However, if the cost of high-quality AI is too high, once the platform quality reaches a certain level, platform profit will decline.

5.3 Comparisons among mechanisms

In this section, we begin by comparing two sequencing mechanisms: AI service first versus sponsored search first. This comparison enables us to recommend an optimal strategy for the platform regarding the order of presenting the AI section and sponsored search. Beyond this, we also extend the analysis to include a third mechanism—our base model—where viewers cannot bypass the AI service to access sponsored search directly.

The following proposition characterizes the optimal sequencing rule when the viewer is given the bypassing option.

Proposition 5. *Within the Paywall mechanisms, the platform prefers AI-First when AI quality is high; the platform prefers AI-Second when AI quality is low.*

Proposition 5 implies that when viewers must pay before using the AI service, the platform’s ranking strategy is essentially determined by the marginal contribution of AI quality to revenue. If the AI quality is high, viewers have high willingness to pay, which implies that placing the AI in the first position can highlight its value and maximize direct revenue generated from the AI service itself. Conversely, if the AI quality is low, placing it into the first position may increase viewers’ friction or dissatisfaction, leading to a decline in paying viewers. In this case, positioning the AI second enables the platform to first engage viewers with higher-quality links. This approach prevents premature exposure of the AI’s limitations while still maintaining its availability as an optional value-added feature. In short, the ranking strategy reflects a trade-off between AI-generated revenue and viewers experience: high-quality AI is prioritized for monetization, while low-quality AI is placed later to mitigate risks.

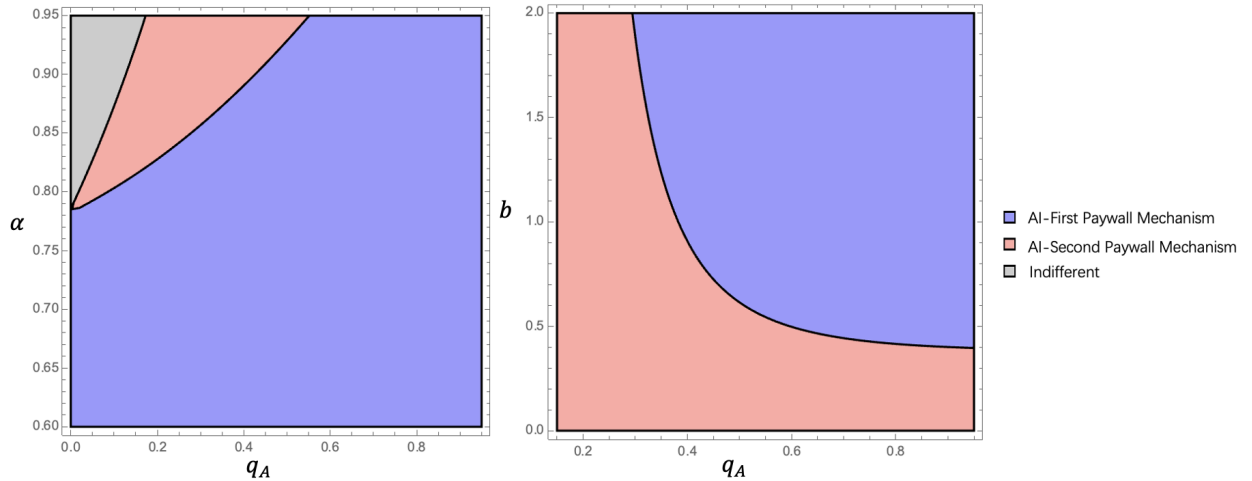


Figure 11: The optimal strategy for the platform under two Paywall mechanisms.

Now we compare all three mechanisms, including our base model wherein the viewer cannot bypass the AI service and go for sponsored search directly. We will consider the Subscriber Intelligence scheme and allow the platform to conduct two-sided pricing. We note that Sponsored Intelligence Scheme is dominated by AI-First Paywall Mechanism, $\Pi^{FAw*} > \Pi^{FA0}$. Accordingly, the following proposition outlines the guiding principles for sequencing decisions and identifies the circumstances under which offering a bypass option is appropriate.

Proposition 6. *Among the three mechanisms (Subscriber Intelligence, AI-First and AI-Second Paywall mechanisms), from the platform's perspective:*

- *AI-First Paywall mechanism dominates when α is low;*
- *Subscriber Intelligence dominates when α and q_A are high;*
- *AI-Second Paywall mechanism dominates when α is high and q_A is in the middle range;*
- *For other regimes, these mechanisms yield the same payoff to the platform.*

Proposition 6 illustrates the distinct pricing mechanisms between scenarios where viewers must pay for AI to access AI part and links and those where they can choose to access links without paying for AI. The fundamental mechanism revolves around how the AI quality influences viewers' willingness to pay, as well as how the marginal payoff for advertisers affects the positioning of the AI component. Within the context of subscriber intelligence, viewers are presented with only two choices: to either pay for the platform's services or to exit. High-quality AI is essential for the platform to attract a larger audience. Without viewers engaging with the platform, the potential business value derived from sponsored links cannot be realized, resulting in a complete lack of profit.

Under the paywall mechanism, CPC emerges as a significant source of revenue. The marginal payoff (α) for the advertiser plays a crucial role in determining the position of the AI component. When the marginal payoff for the advertiser is relatively low, it lacks sufficient motivation to enhance the quality of the sponsored link, leading to the placement of the AI section at the forefront to entice more viewers to pay for AI services. Conversely, when the marginal payoff is relatively high, the advertiser is incentivized to improve link quality, and placing the link at the top can yield greater profits. This scenario gives rise to the dominance of the AI-Second Paywall mechanism.

However, the enhancement of sponsored link quality presents a revenue challenge for the platform, especially when viewers are permitted to bypass the AI section. As a result, the AI-Second Paywall mechanism proves to be less profitable than the Subscriber Intelligence scheme, in which the platform generates revenue directly from viewers while benefiting from their heightened willingness to pay for the sponsored link. Consequently, within the intermediate range of α , the Subscriber Intelligence approach prevails over all paywall mechanisms. Figures 12 and 13 provide different scenarios of the optimal mechanism arrangement.

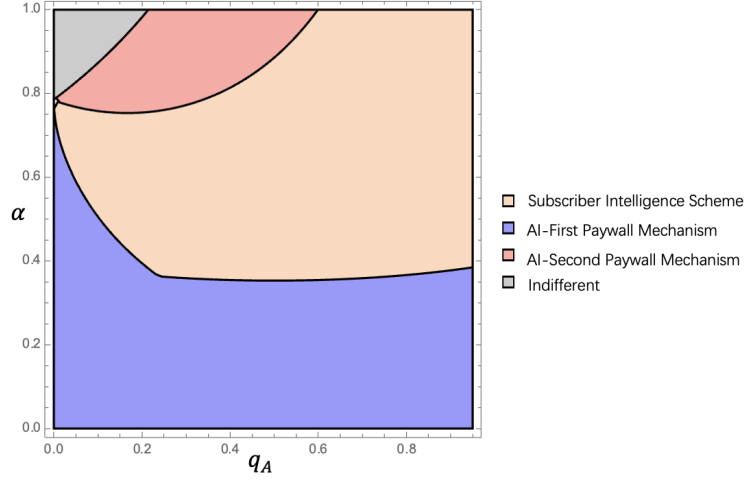


Figure 12: The Optimal Strategy for the Platform $v = 0.7; c = 0.5; \theta_c = 0.55; k = 0.15; b = 1.5$.

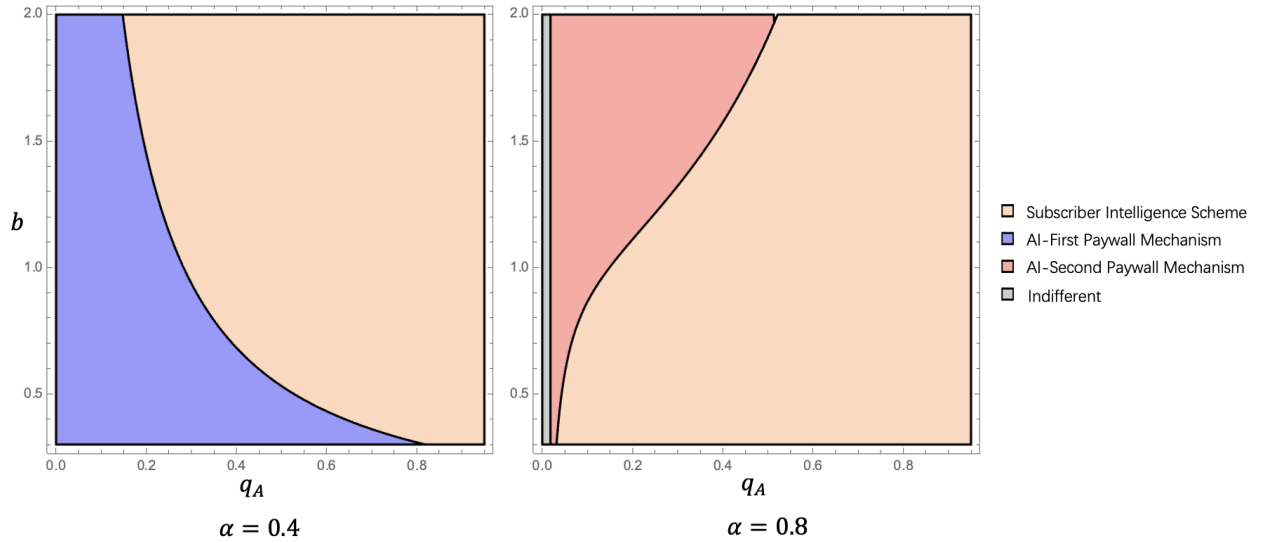


Figure 13: The Optimal Strategy for the Platform $v = 0.7; c = 0.5; \theta_c = 0.55; k = 0.15$.

6 Conclusion

Search engines now confront intensifying competition from artificial intelligence, which increasingly substitutes for conventional query-based search on platforms such as Google and Baidu. In turn, providers have invested heavily in building and integrating AI-driven functionality into their ecosystems. Yet this strategic pivot strains the viability of legacy, sponsored-search monetization. Our study evaluates platform architecture that jointly deploys an AI ser-

vice and sponsored search within a two-sided market, where revenue is extracted from both viewers and advertisers.

Starting from a baseline in which AI precedes sponsored search, we identify how viewer payments for AI access and advertiser payments for placements jointly shape revenue and behavior. We show that the interplay among AI quality, AI pricing, and advertiser quality is non-monotonic: improving the AI does not necessarily raise profits, as higher quality can cannibalize sponsored clicks and weaken advertiser incentives. We also find that paywalled AI does not uniformly dominate free access: when AI quality is low, opening access sustains engagement and advertising value, whereas charging suppresses both.

We then allow a bypass of the AI paywall to assess how access frictions affect viewer participation, advertiser exposure, and total revenue, and we relax the fixed order by permitting AI to follow sponsored search. Our analysis reveals that while giving viewers a paid option can enhance advertiser incentives, optimal sequencing is pivotal; notably, in early AI phases, placing AI after sponsored search can outperform the more common “AI-first” design. Ultimately, the option to bypass the AI service proves to be profitable only when the advertiser’s margin from satisfying the viewer is either exceptionally high or exceptionally low.

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